Department of Computer Science and Business Systems
Academic Year 2023 – 2024(Odd Semester)

Degree, Semester & Branch: B.Tech., III & CSBS
Course Code & Title: AD3351 Design and Analysis of Algorithms
Name of the Faculty member: Ms.K.Usharani, AP/CSBS

Innovative Practice Description

- **Unit / Topic:** Unit 4 / The Maximum-Flow Problem
- **Course Outcome:** CO 4 – Solve the problems using iterative improvement techniques for optimization.
- **Topic Learning Outcome:** Understanding iterative improvement and Able to solve optimization problems.
- **Activity Chosen:** Think Pair Share
- **Justification:**
  1. Think Pair Share is an active learning and also a collaborative learning method which increase the students focus and involvement on learning the topic.
  2. Sharing their knowledge with their teams will improve their understanding in the topic and also their communication skill.
- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**
  1. **Material used :** Activity Sheet to solve Maximum flow problem. Students has to find maximum flow of the given graph. And also they have to find corresponding Minimum cut.
  2. **Prerequisite :** Knowledge on the Maximum flow graph is obtained through previous lecture .
  3. Activity comprises of 3 parts – Think, Pair, Share
  4. **Think** – The students are asked to come up with the solution for the given problem individually.
  5. **Pair** – Students are combined as group of 2 members to discuss their knowledge and also to get clarity on the topic. They are asked to fill the worksheet after discussing with their team members.
  6. **Share** – Based on the solution given by the students, some of the teams were chosen to share their knowledge on the topic to their class students.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PO12	PSO3
CO5	2	1	1	3	3	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO9**	**PO10**	**PO12**	**PSO3**
	2	**1**	**1**	**3**	**3**	**2**	**3**
Justification for correlation	Apply the knowledge of engineering fundamentals like flow conservation concepts.	Identify and analyze complex engineering problems related to graph network flow.	Design solutions for complex engineering problems	Function effectively as an individual and as a member in a team	Communicate effectively on complex engineering activities with engineering community.	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning.	Apply appreciable knowledge of management skills cultured through real time practice.

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***

    Students felt that they can able to understand and get clarity on the topic very easily by sharing their knowledge with their friends.
  - ***Benefit of the practice:***

    Students were actively participated in this activity.

    Students interaction with their peer members and the class during the activity was good and effective.
  - ***Challenges faced in implementation:***

    Some students are hesitated to share their knowledge in the class. And they are motivated to come out of their fear.

References:

- https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improve-your-classroom-discussions1704.html

- https://www.teachingexpertise.com/classroom-ideas/think-pair-share-activity/

- https://www.kent.edu/ctl/think-pair-share